

Nanyang Technological University
AY2024-25, Semester 2
MH1201 (Linear Algebra II)
Midterm Exam

Date: February 26, 2025

Time allowed: 110 minutes

Full name _____

Matriculation Number _____

Tutorial Group _____

INSTRUCTIONS TO CANDIDATES

- (1) **Do not turn over the exam paper until instructed to do so.**
- (2) This midterm exam consists of **four** problems, each consisting of multiple parts.
- (3) Candidates are required to attempt **all four** problems.
- (4) The point allocation for each problem is shown at the top right-hand corner of its page.
- (5) Candidates may write anywhere on the blank pages reserved for showing work.
- (6) **Just one** double-sided A4-sized help-sheet with typed or handwritten notes is permitted.

Problem 1	
Problem 2	
Problem 3	
Problem 4	
Total	

Problem 1**Point allocation:** 25 points

- (a) In each of the following three cases, V is a vector space over $\mathbb{R}$, and W is a subset of V . In each case, determine (with justification) whether or not W is a vector subspace of V .
Note: In each case, one point will be awarded for verifying that $W \neq \emptyset$.

(i) [5 points]

$$V \stackrel{\text{df}}{=} \mathbb{R}^\infty,$$

$$W \stackrel{\text{df}}{=} \{(x_1, x_2, x_3, \dots) \in \mathbb{R}^\infty \mid \forall n \in \mathbb{N} : x_{n+2} = 4x_{n+1} - 2x_n\}.$$

(ii) [5 points]

$$V \stackrel{\text{df}}{=} \mathbb{R}^{3,3},$$

$$W \stackrel{\text{df}}{=} \{\mathbf{A} \in \mathbb{R}^{3,3} \mid \det(\mathbf{A}) = 0\}.$$

(iii) [5 points]

$$V \stackrel{\text{df}}{=} \mathbb{R}^{2,2},$$

$$W \stackrel{\text{df}}{=} \left\{ a \cdot \begin{bmatrix} 2 & -3 \\ 4 & 1 \end{bmatrix} + b \cdot \begin{bmatrix} 0 & 5 \\ 1 & -2 \end{bmatrix} + \begin{bmatrix} 6 & -19 \\ 10 & 7 \end{bmatrix} \in \mathbb{R}^{2,2} \mid a, b \in \mathbb{R} \right\}.$$

- (b) Let V be a vector space over $\mathbb{R}$ (not necessarily finite-dimensional).
 Let $n \in \mathbb{N}$, and let $\mathbf{a}_1, \dots, \mathbf{a}_n, \mathbf{b}_1, \dots, \mathbf{b}_n \in V$ (linear independence is not assumed).
 Suppose that there is an $\mathbf{M} = [\lambda_{i,j}]_{i,j=1}^n \in \mathbb{R}^{n,n}$ with the property that

$$\begin{aligned} \mathbf{b}_1 &= \lambda_{1,1} \cdot \mathbf{a}_1 + \cdots + \lambda_{1,n} \cdot \mathbf{a}_n; \\ \mathbf{b}_2 &= \lambda_{2,1} \cdot \mathbf{a}_1 + \cdots + \lambda_{2,n} \cdot \mathbf{a}_n; \\ \vdots &\quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\ \mathbf{b}_n &= \lambda_{n,1} \cdot \mathbf{a}_1 + \cdots + \lambda_{n,n} \cdot \mathbf{a}_n. \end{aligned}$$

- (i) [4 points] Explain — using the definition of “span” — why

$$\text{Span}(\{\mathbf{b}_1, \dots, \mathbf{b}_n\}) \subseteq \text{Span}(\{\mathbf{a}_1, \dots, \mathbf{a}_n\}).$$

- (ii) [6 points] Show that if $\mathbf{M}$ is invertible, then

$$\text{Span}(\{\mathbf{b}_1, \dots, \mathbf{b}_n\}) = \text{Span}(\{\mathbf{a}_1, \dots, \mathbf{a}_n\}).$$

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Problem 2**Point allocation:** 25 points

Define a function $T : \mathbb{R}^{2,2} \rightarrow \mathbb{R}^{2,2}$ as follows:

$$\forall \mathbf{A} \in \mathbb{R}^{2,2} : \quad T(\mathbf{A}) \stackrel{\text{df}}{=} \mathbf{A} \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \mathbf{A}^\top.$$

Note: The symbol “ $\top$ ” denotes matrix transposition.

- (a) **[5 points]** Show that $T \in \mathcal{L}(\mathbb{R}^{2,2})$.
- (b) **[5 points]** Compute the standard matrix representation of T .
- (c) **[5 points]** Derive a basis for $\text{Null}(T)$.
- (d) **[5 points]** Derive a basis for $\text{Range}(T)$.
- (e) **[5 points]** Determine (with justification) whether or not $\mathbb{R}^{2,2} = \text{Null}(T) \oplus \text{Range}(T)$.
CAUTION: Your solution cannot be a blind application of the Rank-Nullity Theorem!

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Problem 3**Point allocation:** 25 points

- (a) Let V be a 4-dimensional vector space (over an arbitrary field).

[10 points] Show that there is a $T \in \mathcal{L}(V)$ such that $T^4 = -I_V$ (note the minus sign).

Note: $T^4 \stackrel{\text{df}}{=} T \circ T \circ T \circ T$.

- (b) Let V be a vector space (over an arbitrary field).

Let $P \in \mathcal{L}(V)$ be an *idempotent* linear transformation from V to itself, i.e., $P^2 = P$.

Note: $P^2 \stackrel{\text{df}}{=} P \circ P$.

- (i) [4 points] Show, for all $\mathbf{v} \in V$, that $\mathbf{v} - P(\mathbf{v}) \in \text{Null}(P)$.

- (ii) [7 points] Show that $\text{Null}(P) \cap \text{Range}(P) = \{\mathbf{0}_V\}$.

- (iii) [4 points] Hence, show that $V = \text{Null}(P) \oplus \text{Range}(P)$.

CAUTION: The Rank-Nullity Theorem cannot be applied here!

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Problem 4**Point allocation:** 25 points

- (a) Let V and W be vector spaces over a field $\mathbb{F}$ (not necessarily $\mathbb{R}$ or $\mathbb{C}$).
Define a binary operation $+: (V \times W) \times (V \times W) \rightarrow V \times W$ as follows:

$$\text{For all } \mathbf{v}, \tilde{\mathbf{v}} \in V \text{ and } \mathbf{w}, \tilde{\mathbf{w}} \in W: \quad (\mathbf{v}, \mathbf{w}) + (\tilde{\mathbf{v}}, \tilde{\mathbf{w}}) \stackrel{\text{df}}{=} (\mathbf{v} +_V \tilde{\mathbf{v}}, \mathbf{w} +_W \tilde{\mathbf{w}}).$$

Define a binary operation $\cdot: \mathbb{F} \times (V \times W) \rightarrow V \times W$ as follows:

$$\text{For all } \mathbf{v} \in V, \mathbf{w} \in W, \text{ and } \lambda \in \mathbb{F}: \quad \lambda \cdot (\mathbf{v}, \mathbf{w}) \stackrel{\text{df}}{=} (\lambda \cdot_V \mathbf{v}, \lambda \cdot_W \mathbf{w}).$$

[16 points] Show that $V \times W$ is a vector space over $\mathbb{F}$ when equipped with

- $+$ as vector addition and
- $\cdot$ as scalar multiplication.

Note: All eight axioms of a vector space must be verified.

Important remarks:

- This vector space is known as the external direct sum of V with W .
- For the rest of Problem 4, this vector space shall be denoted by “ $V \oplus_{\text{ext}} W$ ”.

- (b) Suppose that V and W are finite-dimensional, with $\dim(V) = m$ and $\dim(W) = n$.
Let $B = \{\mathbf{b}_1, \dots, \mathbf{b}_m\}$ and $C = \{\mathbf{c}_1, \dots, \mathbf{c}_n\}$ be bases for V and W , respectively.

- (i) **[7 points]** Construct a basis for $V \oplus_{\text{ext}} W$ using $\mathbf{b}_1, \dots, \mathbf{b}_m, \mathbf{c}_1, \dots, \mathbf{c}_n$.

Note: Both linear independence and the spanning property must be verified.

Hint: For all $\mathbf{v} \in V$ and $\mathbf{w} \in W$, we have $(\mathbf{v}, \mathbf{w}) = (\mathbf{v}, \mathbf{0}_W) + (\mathbf{0}_V, \mathbf{w})$.

- (ii) **[2 points]** Determine (with justification) $\dim(V \oplus_{\text{ext}} W)$.

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